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Technoverse Hackathon 2026

MVP Presentation Owlgorithms

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TrialGo

Problem Statement

India's ***clinical trial*** participation is critically **low** - not because patients don't exist, but because no intelligent system exists to connect them to trials

- **No Trial Access** - **95%** of eligible patients never find a relevant trial despite **52,000+ studies on CTRI**
- **Recruitment Delays** - **80%** of trials miss deadlines and **55%** get terminated early due to manual, referral-based recruitment
- **Million Dollar Loss -** Pharma sponsors lose **$600K to $8M per day** with no automated candidate sourcing
- **Blind Coordination** - Coordinators discover **dropouts** only after patients disappear — no monitoring, no early warning

Impact:

Patients wait, pharma spends millions, and breakthrough treatments never reach the market.

Solution Description

TrialGo is an **AI-powered platform** using specialized **agents** to automate clinical trial discovery, enrollment and retention.

- **Patient Discovery Portal** — Patients enter their medical profile and AI surfaces the most relevant trials instantly
- **Automated Recruitment** — AI agents scan **15+** global registries and social platforms, eliminating manual recruitment
- **Intelligent Matchmaking** — AI connects the right patients to the right trials, cutting out the middleman
- **Live Monitoring** — Real-time dashboard flags anomalies and predicts dropouts before they happen

A 12-19 month manual recruitment process decreases to days. Patients discovered proactively. Consent understood. Dropouts predicted before they happen.

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TrialGo

Uniqueness / Innovativeness

TrialGo is the only **end-to-end AI system** that handles discovery, recruitment, matching, consent and retention on a single platform.

Feature	What Exists Today	TrialScope
Patient Discovery	Registry search, doctor referrals	AI-powered matching from medical profile
Candidate Sourcing	Manual outreach, hospital EHR mining	Agents scrape 15+ registries + social platforms
Consent Process	40-page English legal documents	2-page plain language in 9+ Indian languages
Dropout Handling	Reported after patient disappears	ML flags risk early, triggers auto re-engagement
Data Standards	Siloed, incompatible formats	FHIR R4 export, compatible with any EHR/system
Platform Scope	Separate tools for each stakeholder	One unified platform for patient, pharma and coordinator

Business Impact

Quantitative Impacts:

- Recruitment time cut from **12-19 months** to **days**
- Patient reach increases from **5%** → **60%+**
- **$600K–$8M** recovered per day saved
- **55%** trial termination rate - significantly reduced
- **30%** dropout rate - predicted and prevented

Qualitative Impacts on:

- **Patients** - More people access trials that could save their lives
- **Pharma** - Faster recruitment, lower searching costs and fewer terminated trials
- **Coordinators** - Proactive monitoring means cleaner data, safer patients and successful trials
- **Healthcare System** - India's clinical trial contribution grows, attracting more global research and investment

Market Opportunity:

- India clinical trial market: **$2.05B (2024)** → **$3.37B by 2030**
- India patient recruitment services: projected **$424.2M by 2033 (10% CAGR)**

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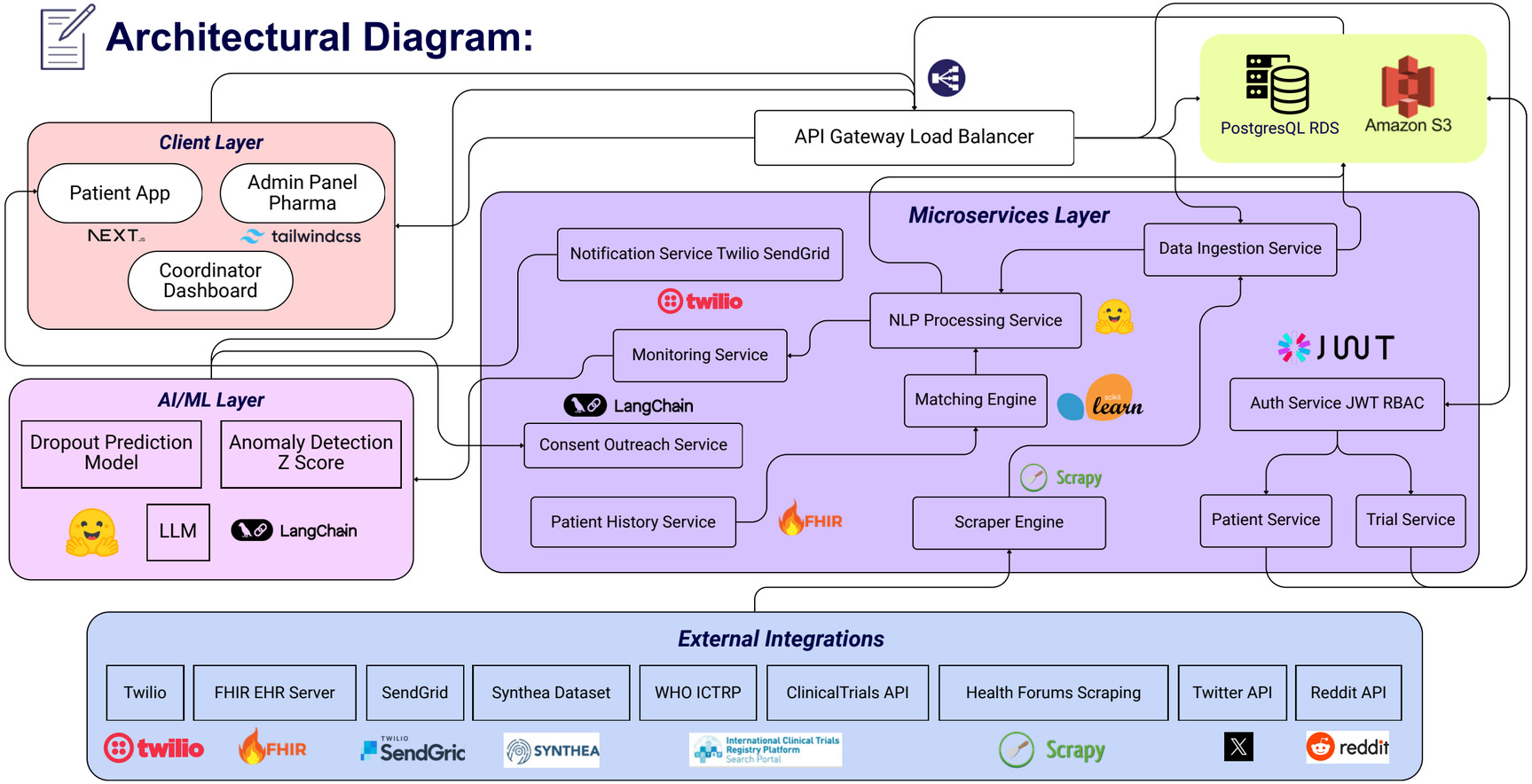
TrialGo

Technical Design and architecture

Architectural Diagram:



Technologies & AI Usage:

- Next.js + FastAPI + PostgreSQL + Celery/Redis
- AWS S3 for secure storage
- GenAI (LangChain): NLP, consent simplification, chatbot
- Multi-LLM: Google (language), Groq (matching), OpenAI (fraud)
- ML: dropout prediction, anomaly detection

Architecture & Uniqueness:

- 3 portals → FastAPI → DB + 12-agent pipeline
- End-to-end flow: discovery → enrollment → retention
- Modular AI agents (scalable & flexible)
- Multi-LLM routing (cost + speed optimized)
- FHIR-native + real-time monitoring (WebSockets)

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** Use additional slides to include other relevant details, as required.

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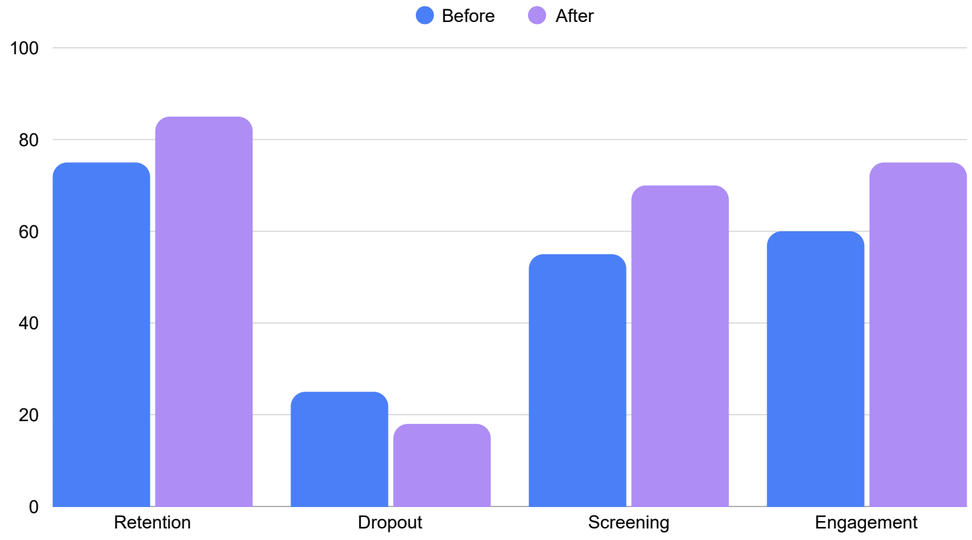
Business Model

	Current Industry	With TrialGo	How It's Reduced
Cost per patient	₹3.5L	₹1.2L	AI-based patient matching reduces irrelevant outreach & screening waste
Total cost (500 patients)	₹17.5 Cr	₹7.5 Cr	Lower per-patient cost + fewer dropouts
Recruitment cost share	25% of trial budget	18%	Direct patient discovery (social + data) reduces marketing spend
Cost overruns (delays)	+25%	+10%	Faster recruitment via automated pipelines



Quality of Care:

- Retention ↑: **75% → 85%**
- Dropout ↓: **25% → 18%**
- Screening ↑: **55% → 70%**
- Recruitment time ↓: **9 → 6 months**
- Engagement ↑: **60% → 75%**

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Scalable / Reusable

Scalability:

- **Stateless FastAPI backend** → easy horizontal scaling
- **Celery workers** scale independently per workload
- **Redis** for caching & rate limiting
- **PostgreSQL** supports scaling (replicas, isolation)
- Docker-based → Kubernetes-ready

Reusability:

- Disease-agnostic → configurable via DB (no code changes)
- Easy region expansion → add registries + language configs
- Plug-and-play data sources (social platforms)
- White-label ready (UI theming)
- API-first → integrates with existing healthcare systems

Roadmap

PHASE 1
MVP Development
- Build core platform (3 portals + backend + AI agents)
- Basic trial matching, onboarding, consent

PHASE 2
Pilot & Validation
- Partner with 1–2 hospitals/pharma
- Run small trials, validate matching & retention

PHASE 3
Scale AI & Integrations
- Add more registries, languages, social sources
- Improve ML models (dropout, anomaly detection)

PHASE 4
Enterprise Readiness
- Security, compliance (FHIR, data privacy)
- API integrations with CTMS/EHR systems

PHASE 5
Market Expansion
- Scale to multiple pharma clients & places
- Extend to adjacent use cases (vaccines, insurance, registries)

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Our Technoverse Journey

The event started beautifully: meeting incredible people, trading ideas, and just enjoying the build. The vibes were truly immaculate. Then came the midway reality check. While looking up the competitors we found out and realized our only and biggest rival wasn't another student team. But it was Cognizant!

Realizing you’re up against the host in their own arena is a very specific type of humbling. But instead of sweating it, we treated it like an unexpected boss level. We laughed, locked in, and went back to shipping clean tech. We came for the community, stayed for the plot twist, and left knowing we can hold our own against anyone.

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Thank you

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